

CONSTRAINED DECODING FOR LARGE LANGUAGE MODELS: TEST-TIME LOGIT FUSION FOR LOW-RESOURCE LANGUAGE ADAPTATION

TIANSHUAI GAO | MPHIL DATA INTENSIVE SCIENCE | UNIVERSITY OF CAMBRIDGE
Word Count: 954/100 (excluding references)

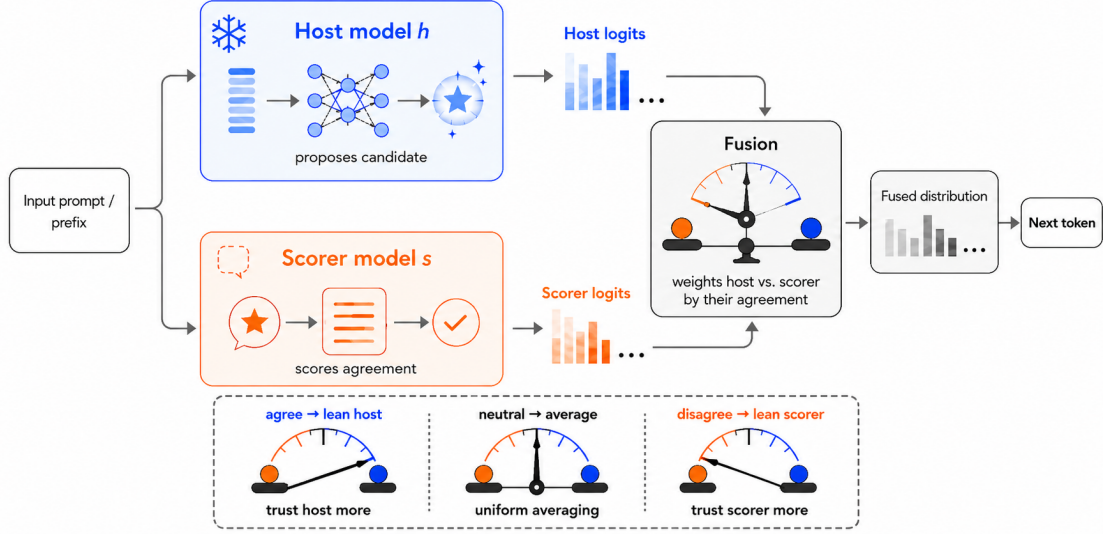


Figure 1: SAF-W at one decoding step. The host proposes its top word, the scorer reports the probability it assigns to that word, and this endorsement sets the mixing weight of the fused distribution.

Motivation & Scientific Justification

Large language models now sit inside search engines, assistants, and public services, but their quality is uneven across languages. Speakers of low-resource languages such as Tibetan, Uyghur, Mongolian, and Kazakh receive far weaker systems, because these languages are scarce in the training data [1]. Closing the gap by training is hard on both fronts. Labeled task data is scarce, since annotators for these languages are few. Continued pretraining at the scale of a 32-billion-parameter model demands compute few such groups can afford. A cheaper route adapts models at decoding time [2]. A small model is continually pretrained on raw text in the target language, at a fraction of the cost. The large model then generates together with it, and the two vote on every word. No parameter of the large model changes. This is constrained decoding with a neural constraint. Where grammar-constrained methods reshape the next-word distribution with a symbolic rule so that outputs parse, here the constraint is a second model, and its distribution corrects the first during generation. Two published methods define this line, Proxy Tuning [3] and TriMix [4], and both were validated on small and mid-size models. This project reproduces both inside one controlled pipeline, evaluates them across three model scales, and asks a question the field had not tested. Does the recipe survive when the large model becomes genuinely strong? The answer is no. We identify the failure, name it the dilution effect, and propose SAF-W (Scorer-Adaptive Fusion with Endorsement Weighting), a simpler method built to avoid it.

Methodology

All experiments run on MiLiC-Eval [5], a benchmark of six tasks in the four languages above, from reading comprehension and response selection to translation in both directions, title generation, and mathematical reasoning. The large models are Qwen2.5 [6] at 7B, 14B, and 32B, and the small model is a

1.5B language-adapted checkpoint. We rebuilt both baselines inside the released TriMix framework rather than reimplementing them, keeping the original caching, batching, stopping rules, and prompts. Every method then runs through one shared generation loop and one shared evaluator, so the comparison isolates the fusion rule and nothing else. Decoding is greedy and deterministic, with fixed prompts, so every reported number is reproducible. Prior fusion methods blend the models with weights fixed once per task, and the same mixture applies to every word. SAF-W instead decides at every word (Figure 1). One model is the host and proposes its most likely word. The other model is the scorer and reports the probability it assigns to that proposal. This probability is the endorsement, and it becomes the mixing weight directly. When the scorer agrees, the host keeps control. When the scorer rejects the word, weight shifts to the scorer. The weight needs no training and no tuning. The only configuration is one bit per task and language, namely the choice of host. We fix it on held-out development data before touching the test set, and the chosen host matches the best host on test in 53 of 60 cells. We also report a per-example oracle that picks the better host with knowledge of the test label. It is not deployable, but it marks the ceiling that any selection rule can reach. We then stress the finding three ways, on a second model family (Gemma-3 [7]), on code-switched translation [8] with Llama-2 and Mixtral hosts, and with a token-level analysis of the weight itself.

Key Findings

The dilution effect is real. At 7B and 14B both prior methods beat the large model alone. At 32B TriMix falls 3.8% below it. A fixed mixture injects a weaker partner into every word, and the stronger the large model, the more this costs. SAF-W reverses the trend. It scores best overall at every scale, and its advantage over TriMix grows from +2.0% at 7B to +6.5% at 14B and +9.4% at 32B, while running two models against

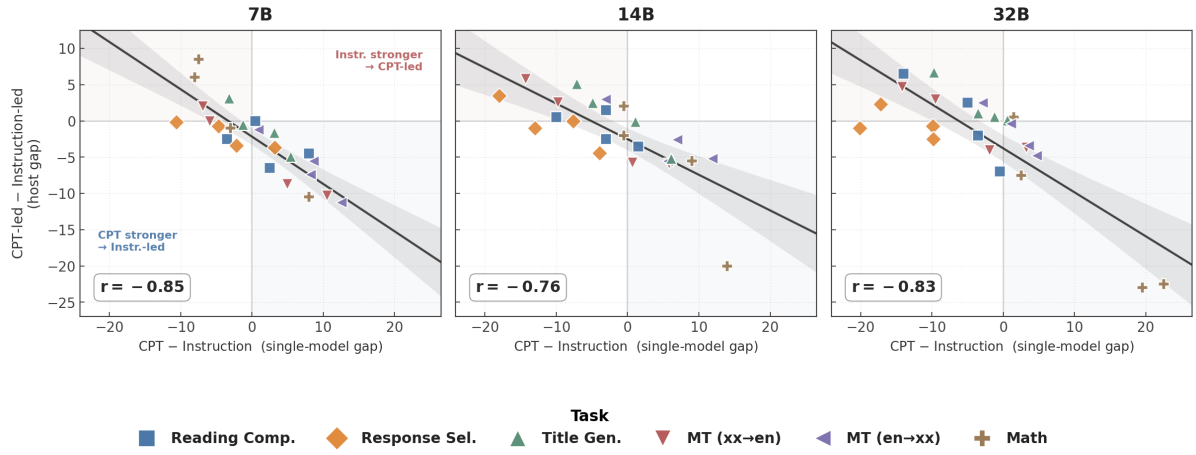


Figure 2: The better host tracks single-model strength, and the pattern holds at every scale. Each point is one task and language. When one model is the stronger standalone system, the other is the better host.

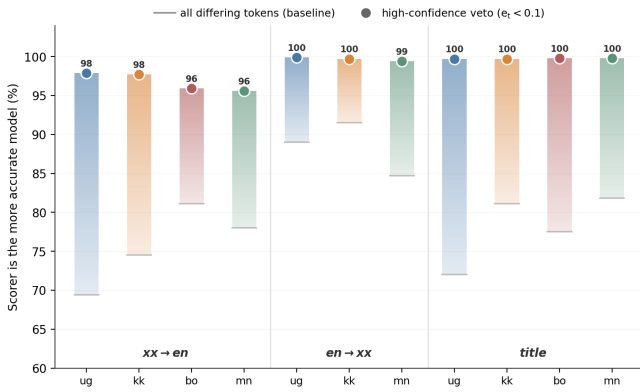


Figure 3: Where the scorer strongly rejects the host’s word, the scorer is the more accurate model 95 to 100% of the time, for every language and direction.

their three. The host choice itself follows one simple law. Whichever model is weaker on its own is the better host, so a development set suffices to pick it (Figure 2). The gains transfer. On Gemma-3, SAF-W beats TriMix at both scales, and the margin again grows with size. On code-switched translation the contrast is sharpest. The Tamil small model rarely leaves Tamil and collapses to 27.5 chrF++. TriMix inherits the failure at 28.7. SAF-W keeps the host in charge and reaches 59.7, above every alternative. A broken partner no longer propagates. The token-level analysis explains the mechanism (Figure 3). On most words the two models agree and the host decodes essentially alone. On the rare words where the scorer strongly rejects the proposal, the scorer is the more accurate model 95 to 100% of the time, across every language. The weight intervenes exactly where intervention pays, and the remaining gap to the oracle traces to one identifiable case, the words where the scorer tolerates the host’s choice while privately preferring a better one. One task resists. Mathematical reasoning needs both abilities in sequence, reading the low-resource question and then computing. With the endorsement active, generation collapses into bare arithmetic over the surface numbers and skips the reasoning chain. SAF-W detects this on development data and decodes math with uniform averaging, a built-in special case that keeps both models active at every word and restores the chain.

Recommendation & Next Steps

Three directions follow. First, the endorsement is one-sided. It reads whether the scorer accepts the host’s word, not whether the scorer holds a better one. A two-sided signal, or a light router reading only the input, could recover part of the two-to-four-point gap to the oracle. Second, the method assumes a shared vocabulary, so extending it across model families needs a tokenizer alignment step before the endorsement can be computed. Third, the scale trend rests on a few points per family, and a denser sweep of host and scorer sizes would pin down how the gain depends on the gap between the two models. For practitioners the recipe is ready now. Take any instruction-tuned model, add a small language-adapted model, and deploy with one bit of configuration.

Research Impact & Conclusion

Language equity in AI should not require the budget of a frontier lab. SAF-W turns adaptation into a decoding-time recipe with no training, two models, and one configuration bit, and its benefit grows with the backbone rather than eroding. It also runs one forward pass fewer per word than the three-model baselines. The findings held across two model families and a code-switching stress test, and the full pipeline is openly released. The stronger tomorrow’s models become, the more this recipe returns.

References

- [1] S. Singh et al. GlobalMMLU: understanding and addressing cultural and linguistic biases in multilingual evaluation. 2025.
- [2] A. Liu et al. DExperts: decoding-time controlled text generation with experts and anti-experts. *ACL*, 2021.
- [3] A. Liu et al. Tuning language models by proxy. *COLM*, 2024.
- [4] C. Zhang et al. TriMix: three-model logit fusion for low-resource languages of China. 2026.
- [5] C. Zhang et al. MiLiC-Eval: a benchmark for minority languages of China. 2025.
- [6] Qwen Team. Qwen2.5 technical report. 2025.
- [7] Gemma Team. Gemma 3 technical report. 2025.
- [8] Y. Kang. CoVoSwitch: machine translation of synthetic code-switched text. *ACL SRW*, 2024.